

Pediatric Bone Age Estimation

with Uncertainty Quantification

Deep learning regression · Post-hoc prediction intervals

RSNA Bone Age Dataset · EfficientNet-B3 · MC Dropout · Conformal Prediction

Why Automate Bone Age Assessment?

CLINICAL CONTEXT

- Pediatric bone age is assessed from hand X-rays to diagnose growth disorders
- Traditional method: radiologist compares X-ray to a reference atlas (Greulich-Pyle)
- **Limitations of the manual approach:**
 - Subjective — significant inter-observer variability documented in literature
 - Time-consuming — 10–15 minutes per image
 - Inconsistent across institutions and experience levels
- **Goal of this project:**
 - Automate prediction with deep learning (EfficientNet-B3)
 - Add calibrated uncertainty intervals — clinicians need to know when not to trust the model

Both methods are post-hoc — applied to an already-trained model without retraining.

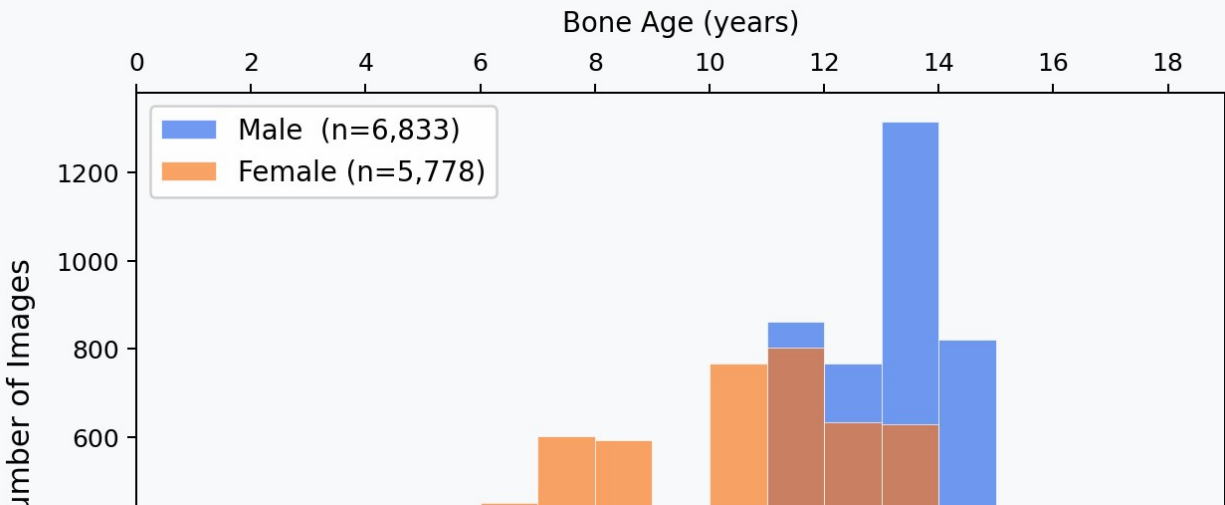
RSNA Pediatric Bone Age Dataset

- Source: RSNA 2017 / Kaggle — publicly available
- 12,611 hand X-ray images**
- Labels per image:
 - boneage — true bone age in months (continuous regression target)
 - male — binary sex indicator (True / False)
- Sex split: 54.2% male / 45.8% female
- Pre-processing:
 - Resize to 300×300, EfficientNet normalization
 - Augmentation: rotation $\pm 20^\circ$, zoom $\pm 10\%$, brightness

Statistic	Value
Min	1 month
Max	228 months (19 yr)
Median	132 months (11 yr)
Mean	127.3 months
Std Dev	41.2 months
5th-95th pct	50 – 186 months

RSNA Pediatric Bone Age Dataset

Age Distribution by Sex



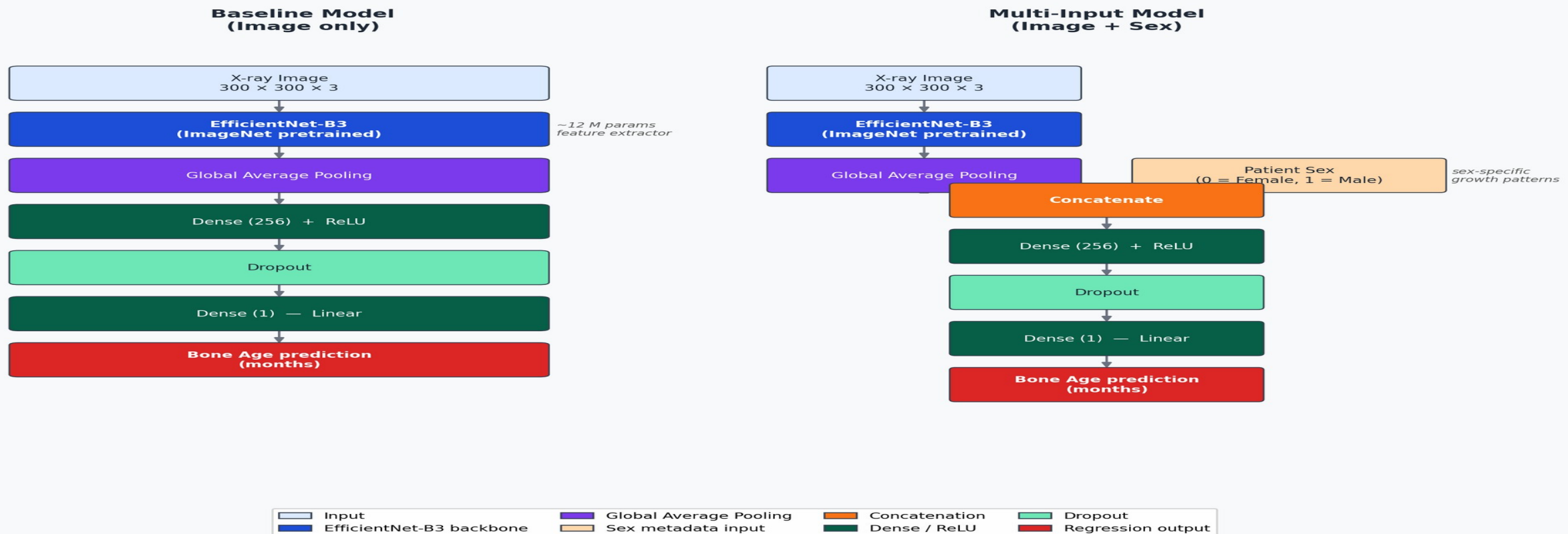
Distribution by Sex (Boxplot)



EfficientNet-B3 Regression Models

- ImageNet-pretrained EfficientNet-B3 · Global Avg Pool → Dense(256)+ReLU → Dropout → Linear output
- Output normalized by max training age; rescaled to months at inference
- Split: 80% train · 10% val (also calibration) · 10% test (seed=42, stratified)
- Training: loss=MAE, Adam, batch=32, early stopping + LR scheduling

EfficientNet-B3 Architecture: Baseline vs Multi-Input



Point Prediction Performance (Regression)

Model	MAE (months)	RMSE (months)	R ²
Baseline — image only	4.47	5.78	0.90
Multi-Input — image + sex	3.82 ↓ 15%	4.98	0.917

- **Adding sex reduces MAE by ~15% (4.47 → 3.82 months)**

- ~80% of predictions fall within ± 6 months of the true age
- ~98% of predictions fall within ± 12 months of the true age
- Error distribution is approximately Gaussian, centred near zero
- Higher errors at extremes: very young ages and older adolescents (sparse data)

These are point predictions — they tell us HOW ACCURATE the model is, but not HOW CONFIDENT it is. That is what UQ addresses next.

From Point Predictions to Trustworthy Intervals

A point prediction answers:

"What is the estimated bone age?"

But a clinician also needs to know:

"How confident should I be in this estimate?"

UQ answers:

- Which predictions are reliable vs uncertain?
- What range covers the true age at a given probability?

Two methods applied:

- Monte Carlo Dropout — model-based, stochastic inference
- Split Conformal Prediction — data-driven, calibration residuals

Prediction	What it tells the clinician
132 months	Model says 11 years — but how reliable is this?
132 ± 16 months (90% CI)	True age is 116–148 months with 90% confidence

UQ Method 1: Monte Carlo Dropout

GAL & GHAHRAMANI, 2016

Standard Dropout: active during training, OFF at inference

MC Dropout: keep dropout ON at inference — $T = 50$ forward passes

- For each test image run the model $T = 50$ times with dropout active
- Each pass randomly zeros different neurons → slightly different prediction
- Collect 50 predictions: $[\hat{y}_1, \hat{y}_2, \dots, \hat{y}_{50}]$
- **Mean of 50 passes = final bone age estimate**
- **Standard deviation of 50 passes = uncertainty estimate**

Interval at confidence level α :

lower = mean - $z(\alpha) \times \text{std}$

upper = mean + $z(\alpha) \times \text{std}$

$z = 1.645$ for 90%, $z = 1.960$ for 95%

Key property:

No separate calibration step needed — uncertainty comes from the model's dropout structure.

Limitation:

No guarantee that interval width reflects actual prediction errors — the model can be overconfident.

UQ Method 2: Split Conformal Prediction

POST-HOC · DISTRIBUTION-FREE · FINITE-SAMPLE COVERAGE GUARANTEE

Step 1 — Calibrate on validation set ($n = 1,261$ images)

Compute residuals: $\text{residuals} = |y_{\text{true}} - y_{\text{pred}}|$ for each image

Step 2 — Compute the conformal quantile

$k = \text{ceil}((n + 1) \times \alpha) \rightarrow \hat{q} = k\text{-th smallest residual}$

Example: $\alpha=0.90$, baseline model $\rightarrow \hat{q} = 20.09$ months

Step 3 — Build test-set intervals

$\text{lower} = \hat{y}_{\text{pred}} - \hat{q} \quad \text{upper} = \hat{y}_{\text{pred}} + \hat{q}$

Interval width = $2 \times \hat{q}$ (same for every image at a given α)

Coverage guarantee: at least α fraction of test samples will have y_{true} inside the interval — under the exchangeability assumption, regardless of model or data distribution.

UQ Metrics: PICP and MPIW

PICP — Prediction Interval Coverage Probability

PICP = fraction of test samples where: $\text{lower} \leq y_{\text{true}} \leq \text{upper}$

- Target: $\text{PICP} \geq \alpha$ (e.g., ≥ 0.90 for a 90% interval)
- $\text{PICP} < \alpha \rightarrow$ intervals too narrow $\rightarrow$ overconfident
- $\text{PICP} \gg \alpha \rightarrow$ intervals too wide $\rightarrow$ over-conservative

MPIW — Mean Prediction Interval Width

MPIW = $\text{mean}(\text{upper} - \text{lower})$ across all test images (months)

- Want this as **SMALL** as possible while still achieving target PICP
- A narrow interval that covers the true value is more informative clinically

The coverage-width tradeoff:

MPIW	PICP	Interpretation
Very wide	High	Safe but uninformative
Very narrow	Low	Precise but unreliable
Narrow	$\geq \alpha \checkmark$	Ideal — calibrated and tight

MC Dropout Results — Overconfident Intervals

MC Dropout Metrics (Baseline vs Multi)

Variant	MAE	RMSE	PICP@0.90	MPIW@0.90	PICP@0.95	MPIW@0.95
Baseline	9.80	12.52	0.710	29.23	0.754	34.83
Multi	8.18	10.77	0.861	35.89	0.904	42.76

- **Baseline severely overconfident: nominal 90% → actual coverage only 71.0%**
- At nominal 95%, baseline coverage is only 75.4% — far below target
- Multi-input improves: 86.1% at $\alpha=0.90$; 90.4% at $\alpha=0.95$ (barely meets one target)
- Widths (29–43 months) are narrower than conformal — but coverage targets are missed
- Multi has wider MC Dropout intervals than Baseline despite better accuracy:
 - Higher predictive std (10.9 vs 8.9 months) → wider Gaussian intervals

MC Dropout uncertainty reflects the model's internal stochasticity, not its actual errors — it cannot self-correct for systematic mis-calibration.

Conformal Prediction Results — Coverage Targets Met

Conformal Prediction Metrics (Baseline vs Multi)

Variant	Alpha	PICP	MPIW	MAE	RMSE
Baseline	0.90	0.894	40.18	9.32	12.02
Multi	0.90	0.903	33.08	7.48	9.94
Baseline	0.95	0.951	49.57	9.32	12.02
Multi	0.95	0.960	43.71	7.48	9.94

- Both variants meet their nominal coverage targets at $\alpha=0.90$ and $\alpha=0.95$

- This is the conformal guarantee working in practice

- Multi-input is strictly better on both dimensions:

- Higher PICP AND narrower intervals than baseline at every alpha level

- Narrower multi intervals explained by calibration: better point predictions →

- smaller validation residuals → smaller $\hat{q}$ (20.09 → 16.54 months at $\alpha=0.90$) → tighter intervals

- Width increases with α (40.18 → 49.57 for baseline) — expected behaviour

$\hat{q}$ values (conformal threshold from validation residuals):

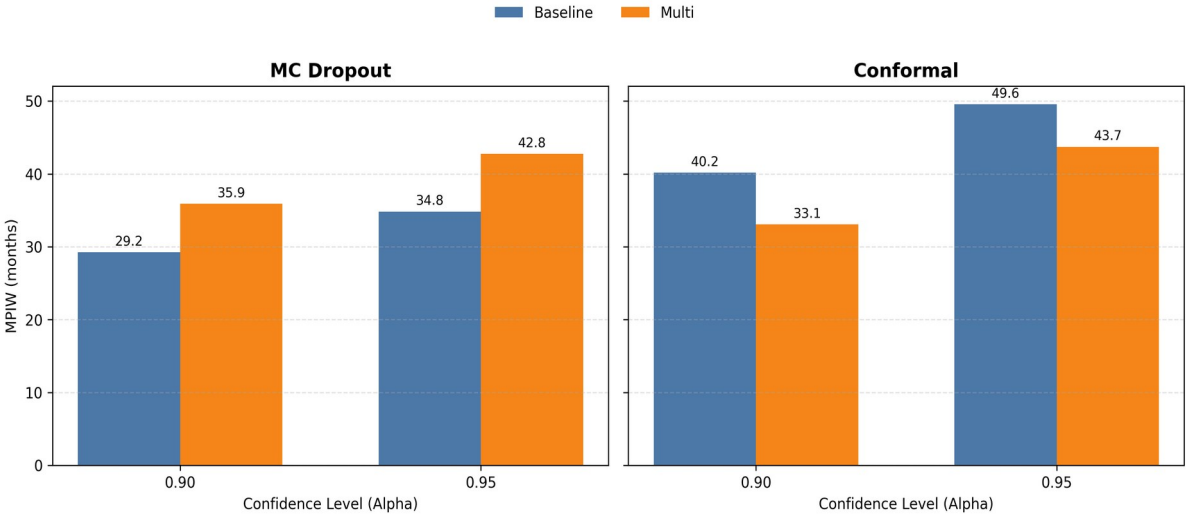
Baseline $\alpha=0.90$: 20.09 mo · $\alpha=0.95$: 24.78 mo Multi $\alpha=0.90$: 16.54 mo · $\alpha=0.95$: 21.86 mo (MPIW = $2 \times \hat{q}$, symmetric intervals)

Combined Comparison: Method × Variant × Coverage Level

UQ Method Comparison (MC Dropout vs Conformal)

Method	Variant	Alpha	PICP	MPIW	MAE	RMSE
Conformal	Baseline	0.90	0.894	40.18	9.32	12.02
Conformal	Multi	0.90	0.903	33.08	7.48	9.94
Conformal	Baseline	0.95	0.951	49.57	9.32	12.02
Conformal	Multi	0.95	0.960	43.71	7.48	9.94
MC Dropout	Baseline	0.90	0.710	29.23	9.80	12.52
MC Dropout	Multi	0.90	0.861	35.89	8.18	10.77
MC Dropout	Baseline	0.95	0.754	34.83	9.80	12.52
MC Dropout	Multi	0.95	0.904	42.76	8.18	10.77

MPIW Comparison by Method, Variant, and Confidence Level



- MC Dropout (left chart): Multi has WIDER intervals than Baseline
 - Counter-intuitive — but multi has higher predictive std (10.9 vs 8.9 mo) → wider Gaussian intervals
- Conformal (right chart): Multi has NARROWER intervals than Baseline
 - Better predictions → smaller residuals → smaller $\hat{q}$ → tighter intervals

Method	Variant	PICP @ 0.90	MPIW @ 0.90	Verdict
MC Dropout	Baseline	0.710	29.23	✗ Under-covers
MC Dropout	Multi	0.861	35.89	✗ Under-covers
Conformal	Baseline	0.894	40.18	✓ Meets target
Conformal	Multi	0.903	33.08	✓ Best overall

Key Takeaways

1 Adding sex metadata consistently improves the model.

MAE drops ~15% and conformal intervals narrow by ~7 months at $\alpha=0.90$.

2 Conformal prediction reliably achieves its stated coverage.

Both variants meet targets at $\alpha=0.90$ and $\alpha=0.95$ — provable under exchangeability.

3 MC Dropout is overconfident, especially for the baseline model.

At nominal 95%, actual baseline coverage is only 75.4%. Intervals are narrower but miss the target.

4 Best combination: Conformal + Multi-input.

90.3% coverage with 33.1-month intervals at $\alpha=0.90$ — reliably calibrated and tight.

5 MC Dropout is still useful as a relative uncertainty signal.

Even if absolute coverage is off, high std → flag uncertain images for clinical review.

6 Conformal intervals are fixed-width per alpha.

Every image gets the same interval — not adaptive to image difficulty. Main limitation.

Limitations and Future Work

CURRENT LIMITATIONS

- No hand segmentation — background noise and cropping artifacts can degrade predictions
- Only sex used as metadata — height, weight, ethnicity, pubertal stage could help
- Conformal intervals are symmetric and non-adaptive — same width for every image at a given α
- Higher errors at extreme ages (infants, older adolescents) are not specifically addressed

FUTURE DIRECTIONS

- Segmentation pre-processing: U-Net to isolate hand region before EfficientNet
- Richer metadata: height, weight, clinical variables as additional model inputs
- Adaptive conformal prediction: Mondrian / locally-weighted methods for image-specific interval widths
- Hybrid architectures: CNN + Transformer (ViT) for richer feature extraction
- Deployment: FastAPI REST endpoint for real-world clinical integration